

367-1-4361

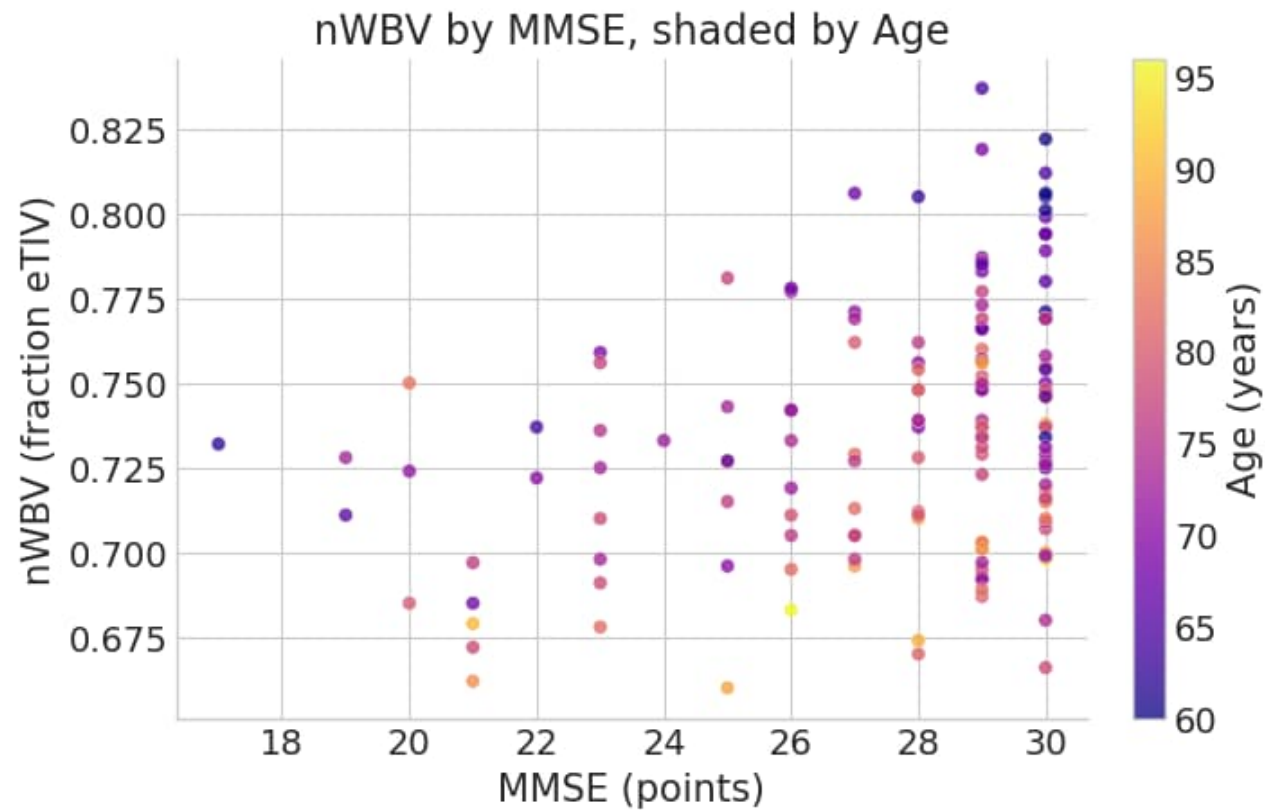
Categorical Data

Opher Donchin

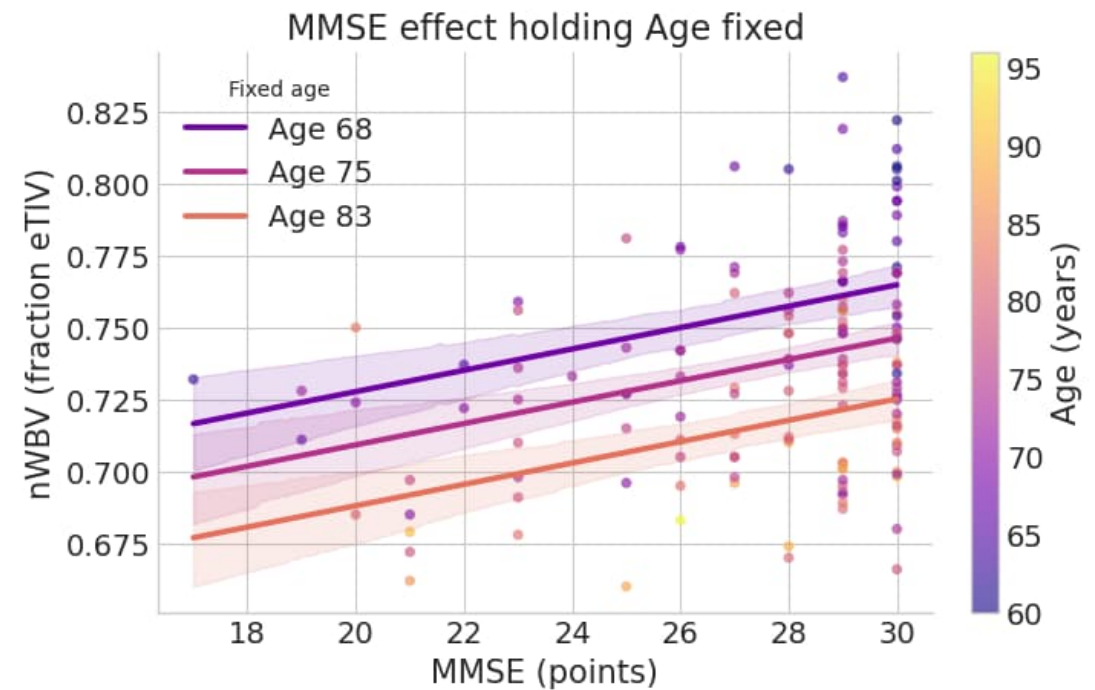
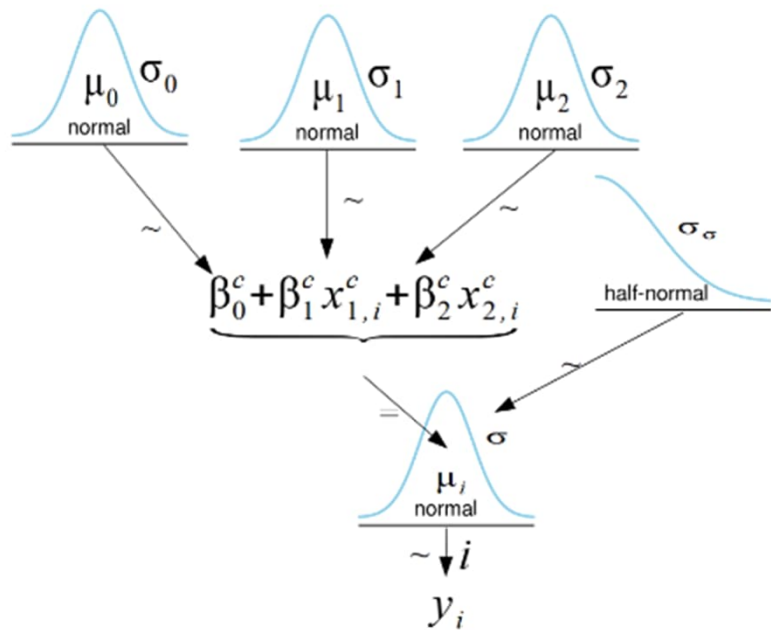
9.1 Regression notation

A standard way of describing regression models

Regression explains dependent with independent variables



Regression models express relationships as slope parameters



Wilkinson notation is a grammar for regression models

- The models we have used so far are strictly linear
 - $y \sim x_1 + x_2$ means
 - $y_i \sim N(\mu_i, \sigma)$ and $\mu_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2}$
- Also called *formula notation*

Models we considered last class

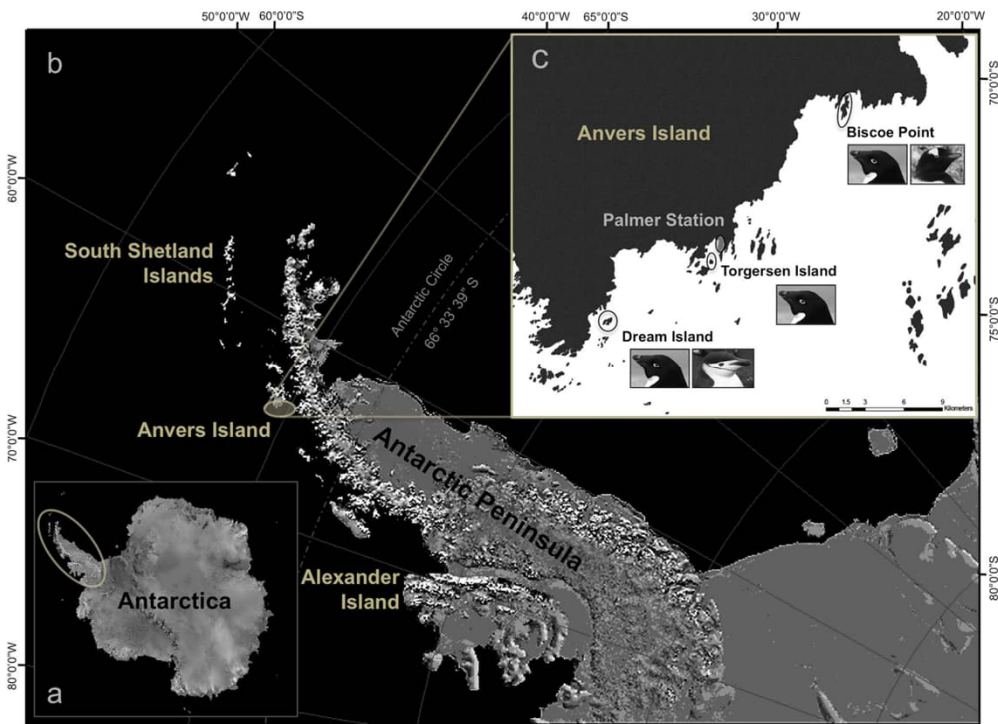
$$nWBV \sim Age$$

$$nWBV \sim MMSE$$

$$nWBV \sim Age + MMSE$$

We explore Wilkinson notation using the Penguin dataset

- Ecological Sexual Dimorphism and Environmental Variability within a Community of Antarctic Penguins (Genus *Pygoscelis*).
 - Gorman, K. B., Williams, T. D. & Fraser, W. R. , *PLOS ONE* 9, e90081 (2014).



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Adélie



Chinstrap



Gentoo

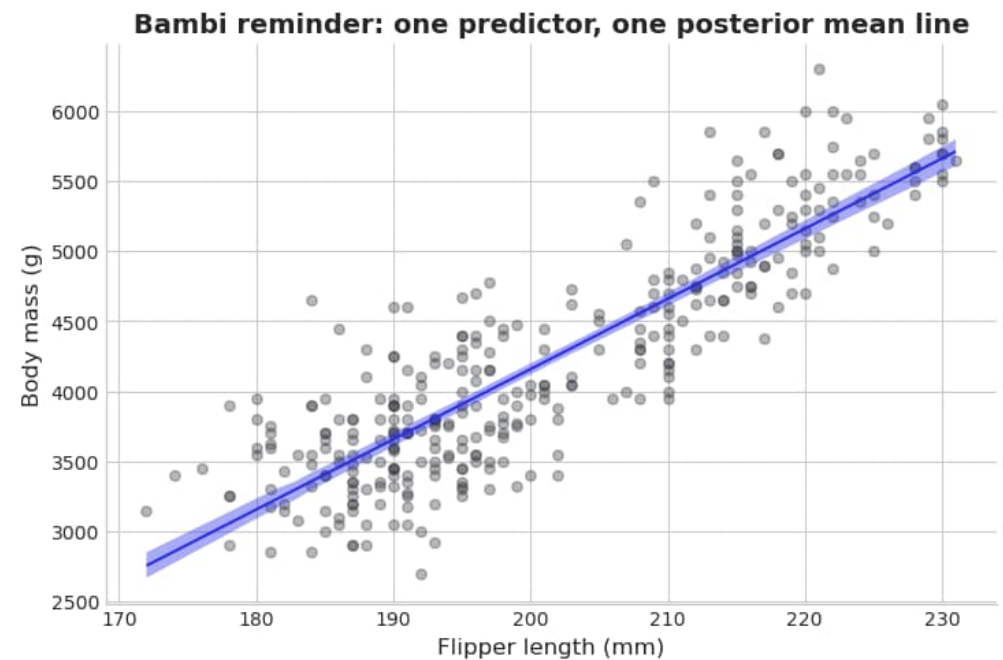


We can regress body mass on flipper length

- Variables we will use include:

- Body mass
- Flipper length
- Bill length
- Species
- Sex

```
bmb.Model(  
  "body_mass_g ~ flipper_length_mm",  
  data=penguins,  
)
```

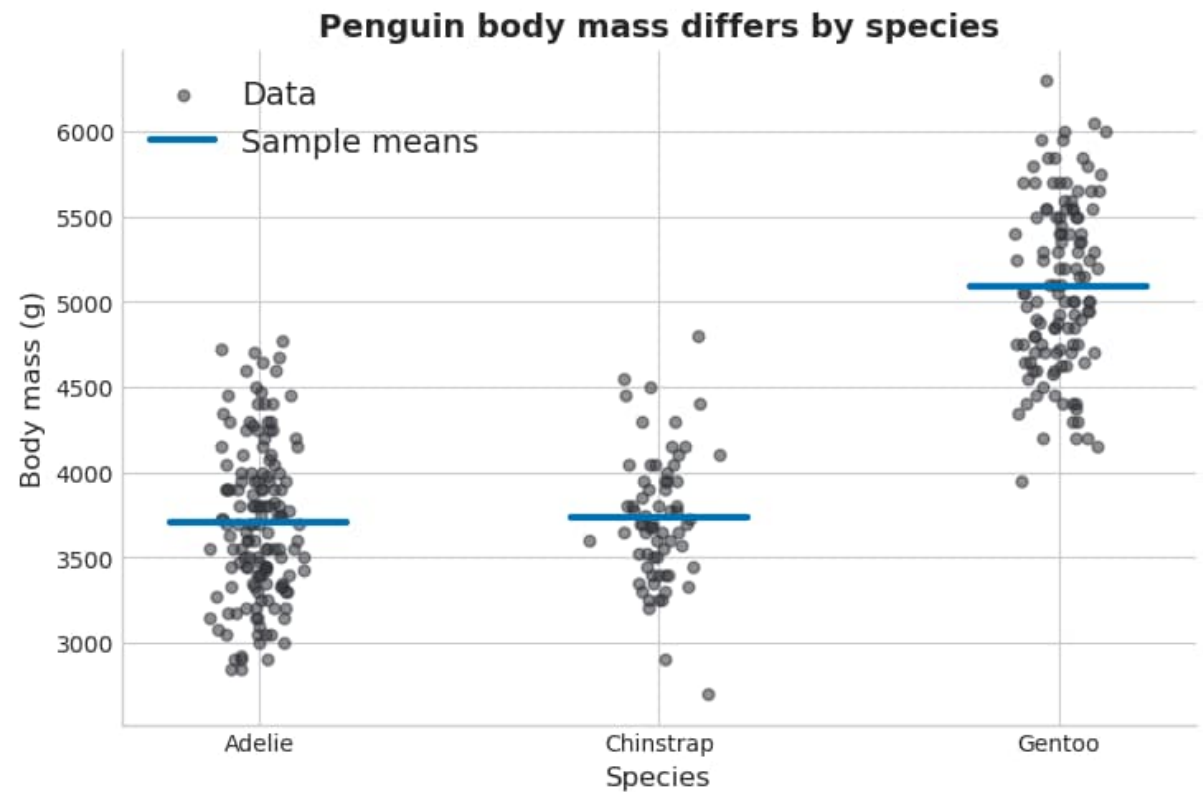


9.2 Categorical predictors

Explaining the dependent variable using groups or categories

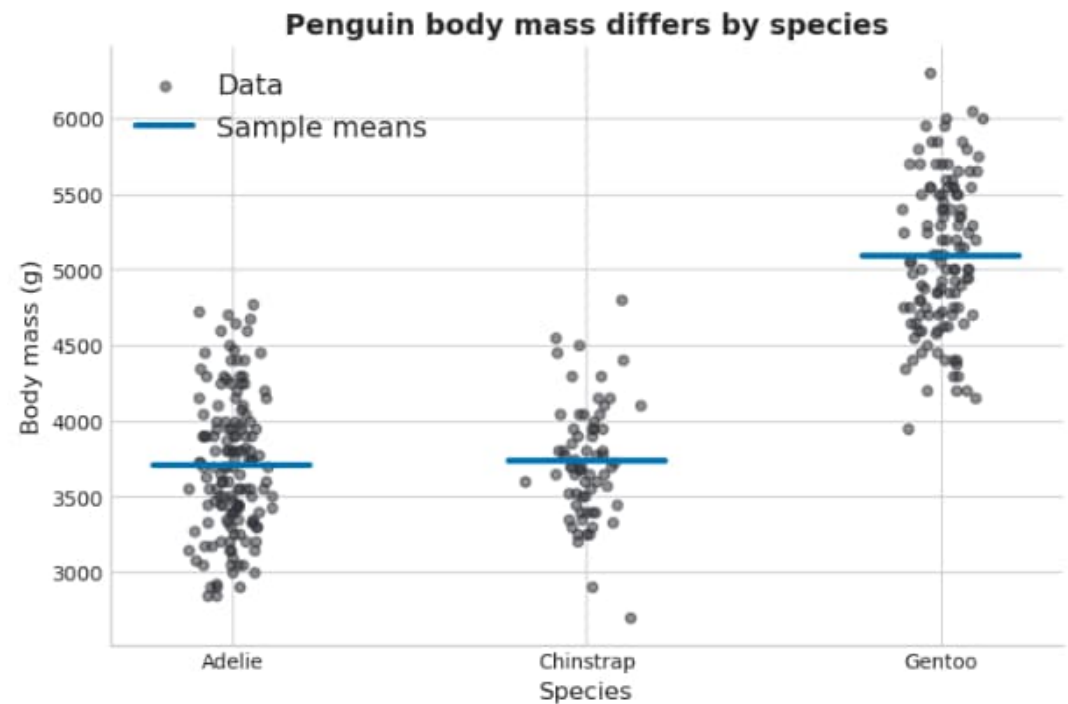
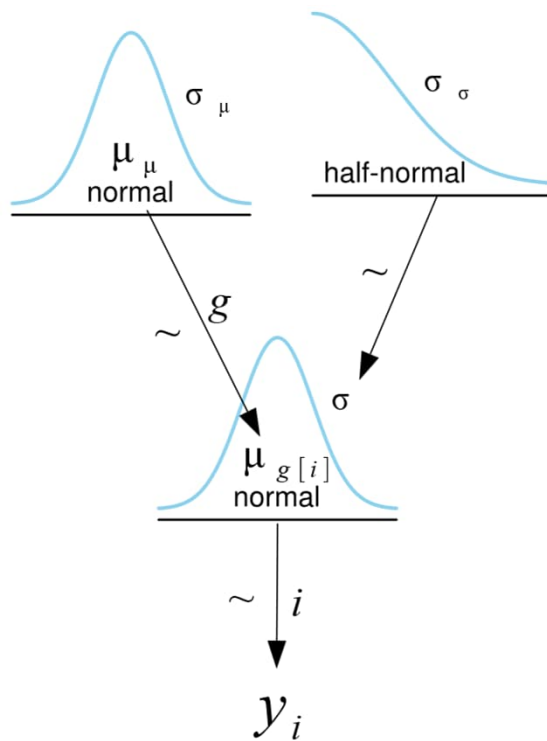
Penguin size varies by species

- Normal models have already been covered
- View them again
 - From a regression framework



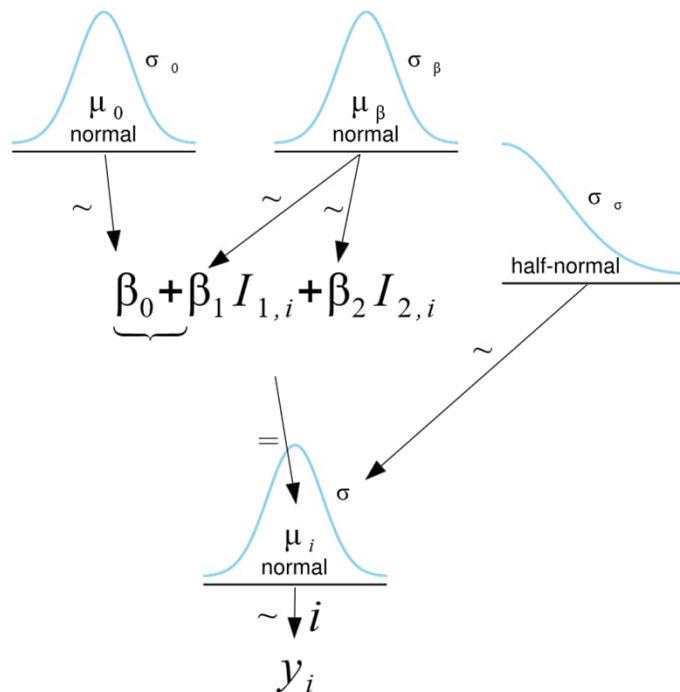
Can use an unpooled normal model

- A different mean for each group



We can write categorical variables as a regression

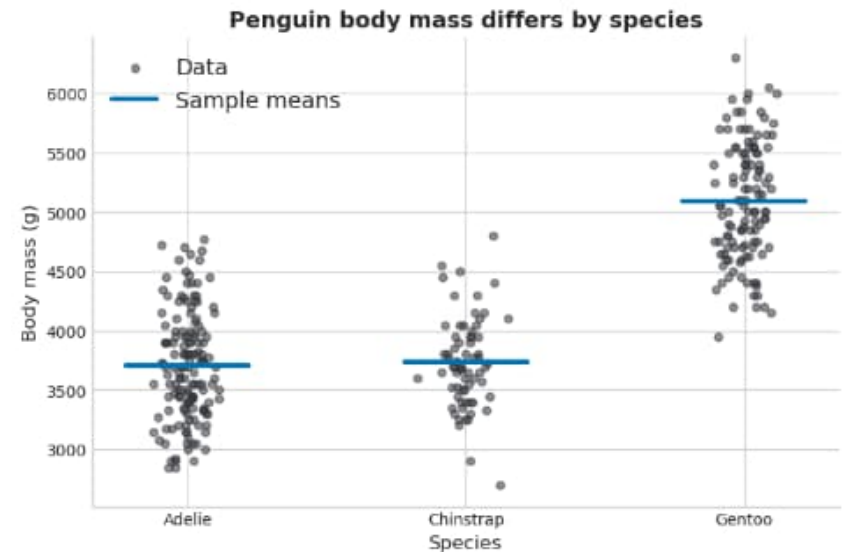
- Use indicator variables
- Get a different combination of parameters for each group
- In Wilkinson notation
 - `body_mass ~ species`



$$I_{1,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases}$$

$$I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

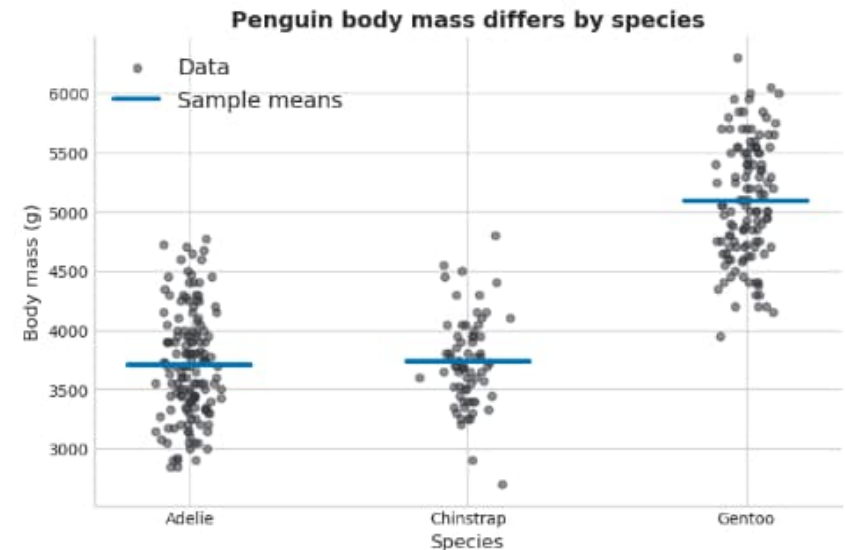
$$\mu_i = \begin{cases} \beta_0 & \text{if species}_i = \text{Adelie} \\ \beta_0 + \beta_1 & \text{if species}_i = \text{Chinstrap} \\ \beta_0 + \beta_2 & \text{if species}_i = \text{Gentoo} \end{cases}$$



Parameters estimate means for each species

- β_0 estimates the mean of the Adelie penguins
 - $y_i \sim N(\beta_0, \sigma)$ if species_{*i*} is Adelie
- β_1 estimates the difference between Chinstrap and Adelie penguins
 - $y_i \sim N(\beta_0 + \beta_1, \sigma)$ if species_{*i*} is Chinstrap

$$\mu_i = \begin{cases} \beta_0 & \text{if species}_i = \text{Adelie} \\ \beta_0 + \beta_1 & \text{if species}_i = \text{Chinstrap} \\ \beta_0 + \beta_2 & \text{if species}_i = \text{Gentoo} \end{cases}$$



Indicator variables coding categories is dummy coding

- Different types of dummy coding
 - Reference: one group represented by β_0
 - Effects
 - Each dummy variables adds to 0 across groups
 - β_0 is the grand mean of all groups
 - Cell means: no special intercept, one indicator variable for each group
 - β_i is the mean of group i

Reference

$$I_{1,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases}$$

$$I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

Effects

$$I_{1,i} = \begin{cases} 1 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ -1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

$$I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ -1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

Cell means

$$I_{1,i} = \begin{cases} 1 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases}$$

$$I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases}$$

$$I_{3,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

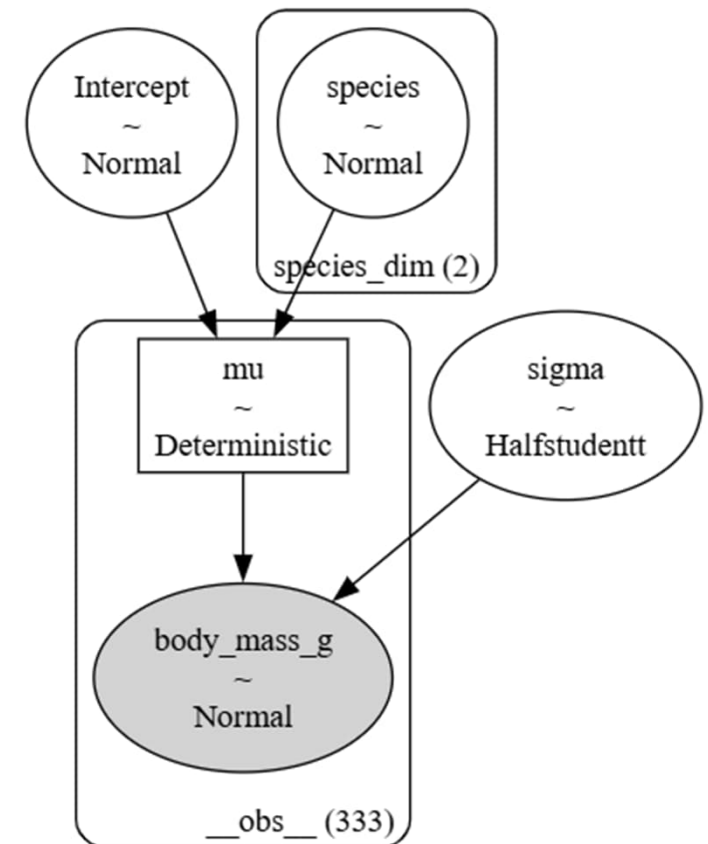
Bambi does dummy coding for categorical variables

- Parameter names are assigned according to coordinate values of dims
- Priors and dummy coding handled automatically

```
bmb.Model(  
  "body_mass_g ~ species",  
  data=penguins_species,  
)
```

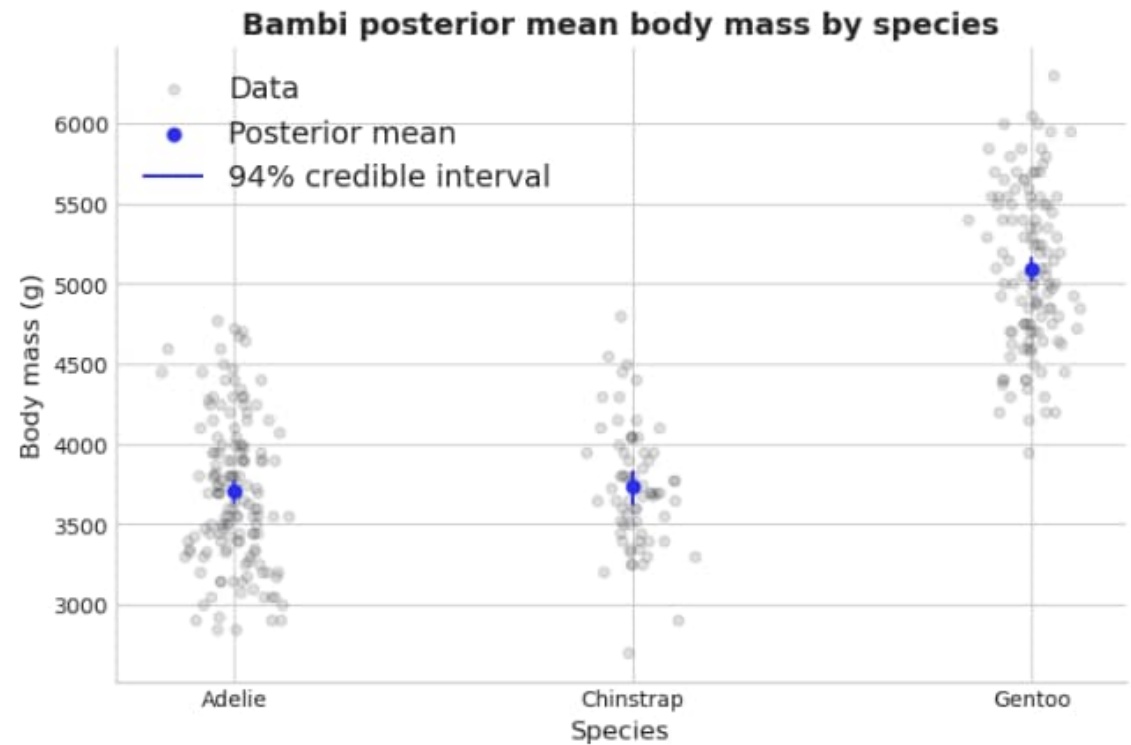
Bambi design-matrix coefficient names: Intercept, species[Chinstrap], species[Gentoo]

	species	Intercept	species[Chinstrap]	species[Gentoo]	expected_mean
0	Adelie	1	0	0	Intercept
1	Chinstrap	1	1	0	Intercept + species[Chinstrap]
2	Gentoo	1	0	1	Intercept + species[Gentoo]



Posterior means are estimated by model

	mean	sd	hdi_3%	hdi_97%
mean_body_mass[Adelie]	3706.960	38.391	3631.540	3776.37
mean_body_mass[Chinstrap]	3732.256	55.827	3624.600	3836.94
mean_body_mass[Gentoo]	5092.106	42.158	5011.514	5167.22



Coefficients are differences from Adelie penguins

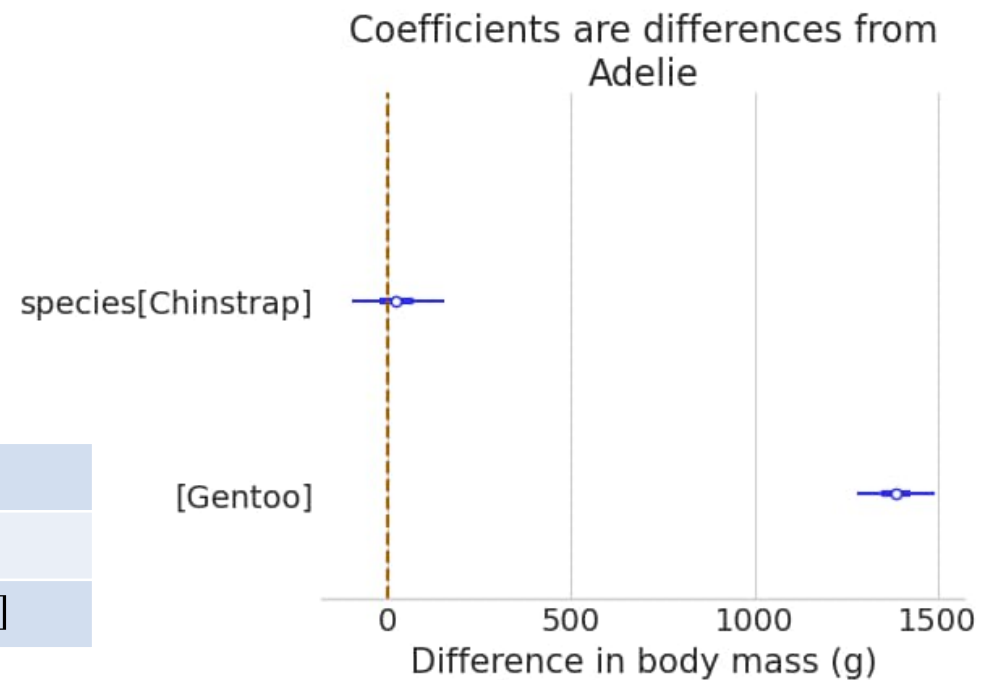
- Gentoo penguins seem bigger

```
coef_summary = az.summary(idata_mass_species, var_names=["species"])
```

	mean	sd	hdi_3%	hdi_97%
species[Chinstrap]	25.296	67.623	-98.350	153.913
species[Gentoo]	1385.146	57.599	1279.879	1491.911

Group differences as coefficient differences

Chinstrap vs Adelie	species[Chinstrap]
Gentoo vs Adelie	species[Gentoo]
Chinstrap vs Gentoo	species[Gentoo] – species[Chinstrap]



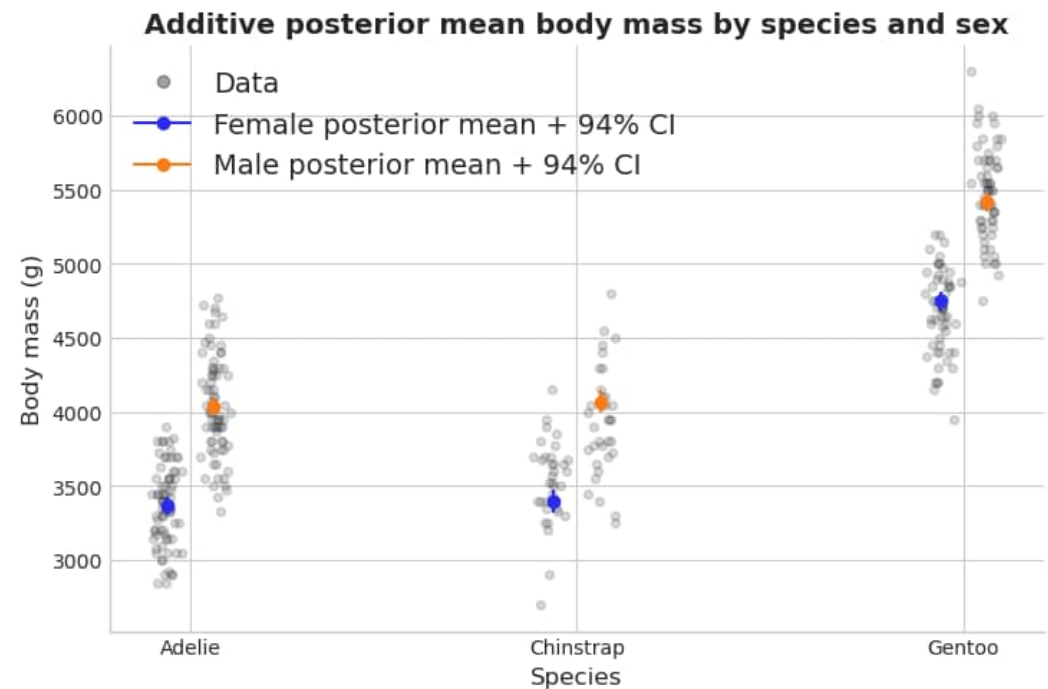
9.3 Two categorical predictors

Extending regression to multiple categorical predictors

We can just as easily fit two categorical predictors

- Both species and sex will be coded with dummy variables

```
bmb.Model(  
  "body_mass_g ~ species + sex",  
  data=penguins_species_sex,  
)
```



We describe categorical regression variables as factors

- Factors
 - A variable in the Wilkinson notation
 - A single categorical variable
 - Like dims in a PyMC model
- Levels
 - A possible value for a factor
 - Like a coordinate in a dim of a PyMC model

The additive model makes strong assumptions

- Sex differences are the same for all species

$$\mu_i = \beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i} \quad I_{\text{sex},i} = \begin{cases} 1 & \text{if male} \\ 0 & \text{if female} \end{cases}$$

Two dummy variables for species

One dummy variable for sex

For the difference between any male penguin i and female penguin j from the same species:

$$\begin{aligned} \mu_i - \mu_j &= \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} \right) - \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,j} \right) \\ &= \beta_{\text{sex}} \quad \text{because } I_{\text{species},k,i} = I_{\text{species},k,j} \text{ for all } k \end{aligned}$$

The assumption is symmetric across factors

- The difference between two species is the same for both genders

For the difference between any two male or two female penguins i and j from different species:

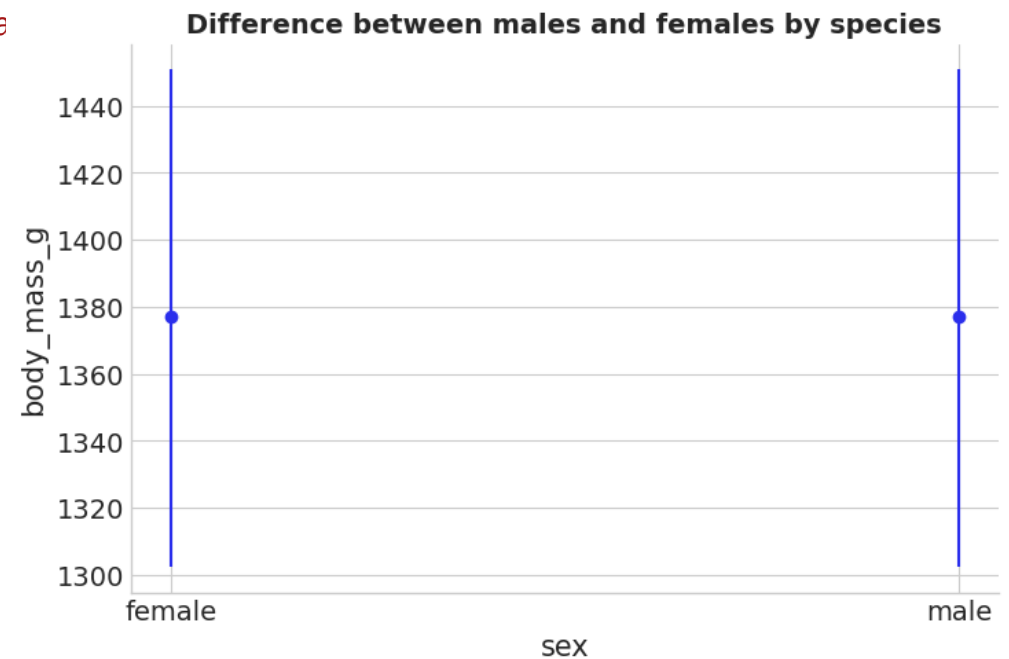
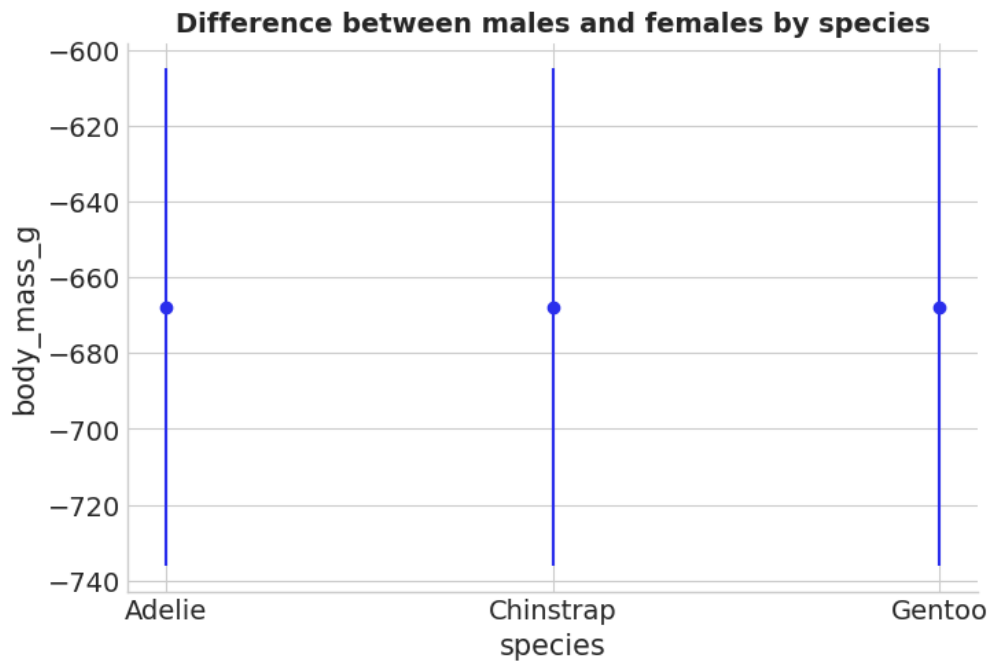
$$\begin{aligned}\mu_i - \mu_j &= \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i} \right) - \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,j} + \beta_{\text{sex}} I_{\text{sex},j} \right) \\ &= \sum_k \beta_{\text{species},k} I_{\text{species},k,i} - \sum_k \beta_{\text{species},k} I_{\text{species},k,j} \text{ because } I_{\text{sex},i} = I_{\text{sex},j}\end{aligned}$$

No interaction between sex and species in the additive model

-

```
bmb.interpret.plot_comparisons(model=model, idata=idata,
                               contrast={"sex": ["male", "female"],
                                         conditional="sex"})

bmb.interpret.plot_comparisons(model=model, idata=idata,
                               contrast={"species": ["Adelie", "Gentoo"],
                                         conditional="species"})
```



9.4 Interactions

Allowing the value of one predictor to change the affect of another

Interactions mean changing slopes

- For two continuous predictors
 - The additive model has a fixed slope for each one
- In an interaction model:
 - the slope of one variable depends on the value of the other
 - Written as an additive model with an interaction term

Additive model $\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Interaction model

$$\mu_i = \beta_0 + (\beta'_{1,0} + \beta'_{1,2} x_2) x_1 + (\beta'_{2,0} + \beta'_{2,1} x_1) x_2$$

$$= \beta_0 + \beta'_{1,0} x_1 + \beta'_{2,0} x_2 + (\beta'_{1,2} + \beta'_{2,1}) x_1 x_2$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

Wilkinson notation can specify interaction terms

- For adding an interaction term use $x_1 : x_2$
- For adding all additive and interaction terms use $x_1 * x_2$

$$\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

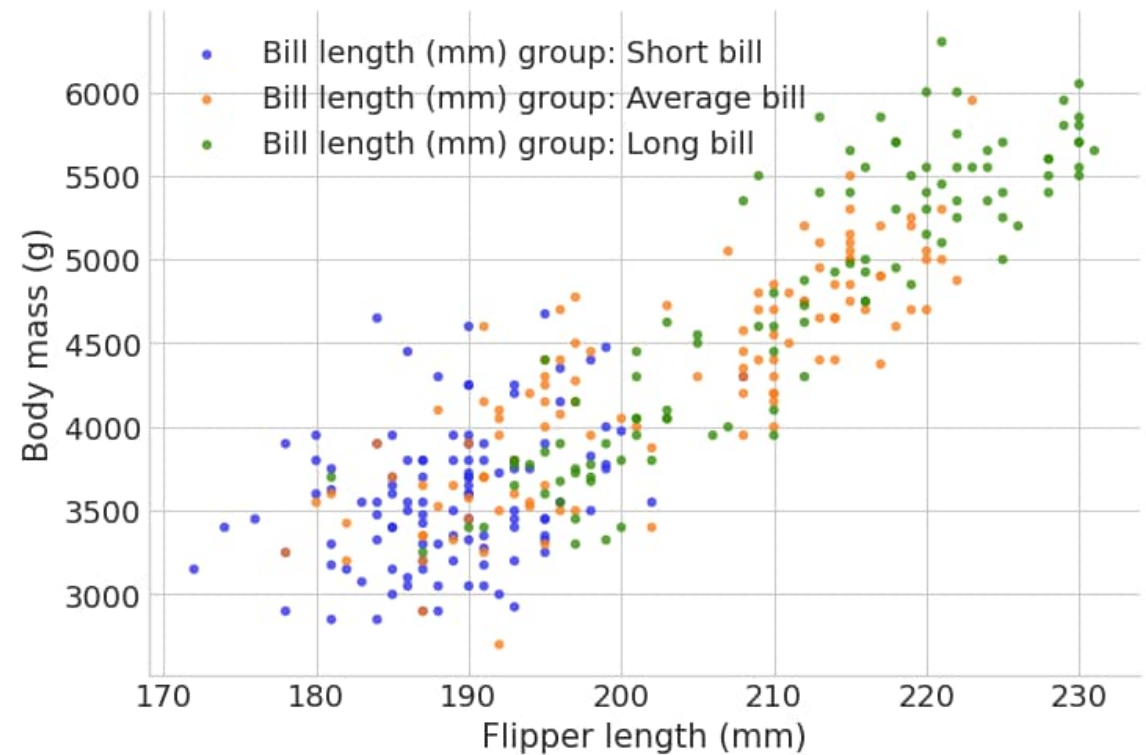
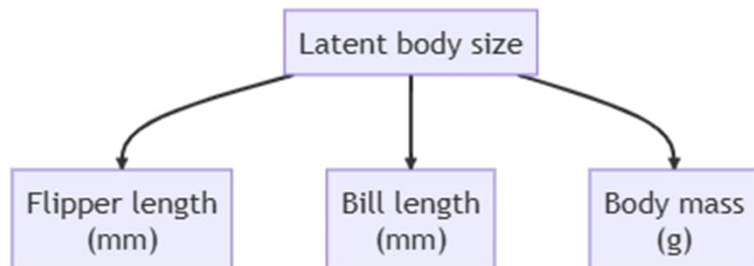
$$y \sim x_1 + x_2 + x_1 : x_2$$

$$y \sim x_1 * x_2$$

Examine how flipper length and bill length predict body mass

- By eye
 - Flipper length predicts body mass
 - Bill length reflect flipper length and body mass

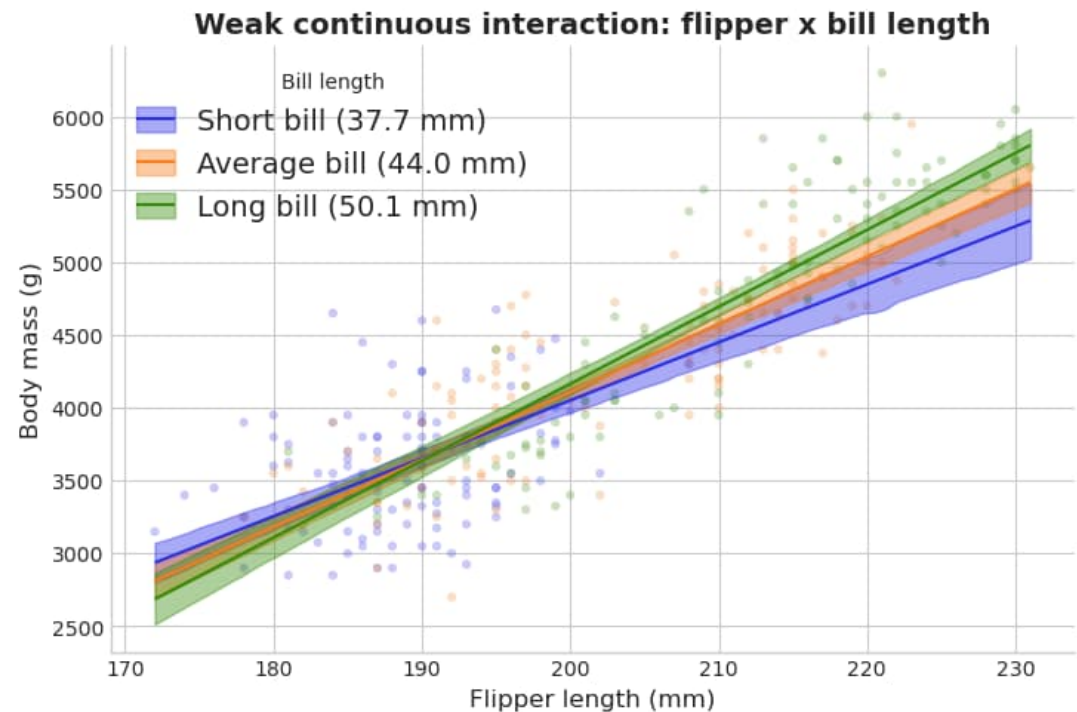
A possible DAG



Fit a Bambi model to see the changing slope

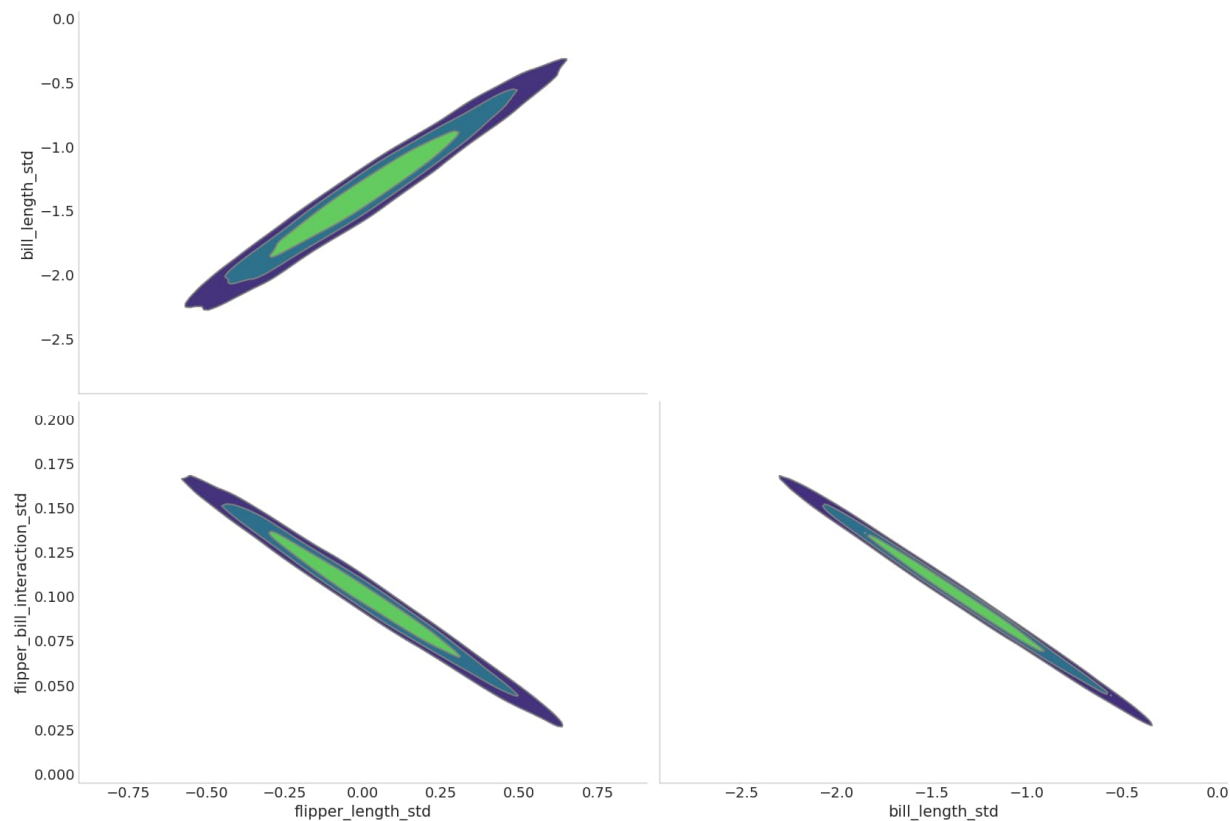
- Specify the interaction model
 - And automatically get changing slope

```
bmb.Model(  
    "body_mass_g ~ flipper_length_mm * bill_length_mm",  
    data=penguins_cont,  
)  
  
bmb.interpret.plot_predictions(  
    model, idata,  
    conditional={  
        "flipper_length_mm": np.linspace(  
            penguins_cont["flipper_length_mm"].min(),  
            penguins_cont["flipper_length_mm"].max(),  
            80,  
        ),  
        "bill_length_mm": bill_level_values,  
    }  
)
```



Highly correlated regression coefficients are hard to interpret

- Regressing on highly correlated predictors
 - Just creates uncertainty



9.5 Interacting categorical variables

How do dummy variables work when categorical variables interact?

We can test for an interaction model in categorical variables

- In Wilkinson notation “0 +” means no intercept
 - Switches categorical variables to cell means coding
 - Dummy variable for each combination of species and sex

```
bmb.Model(  
  "body_mass_g ~ 0 + species:sex",  
  data=penguins_cell,  
)
```

Sampled parameters

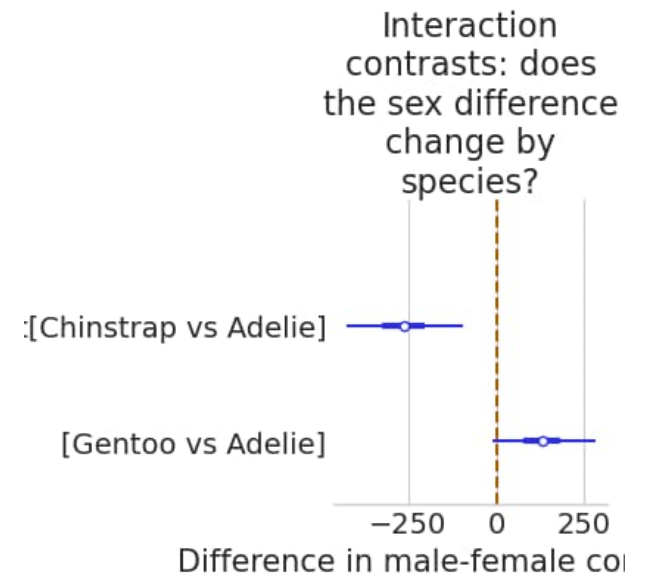
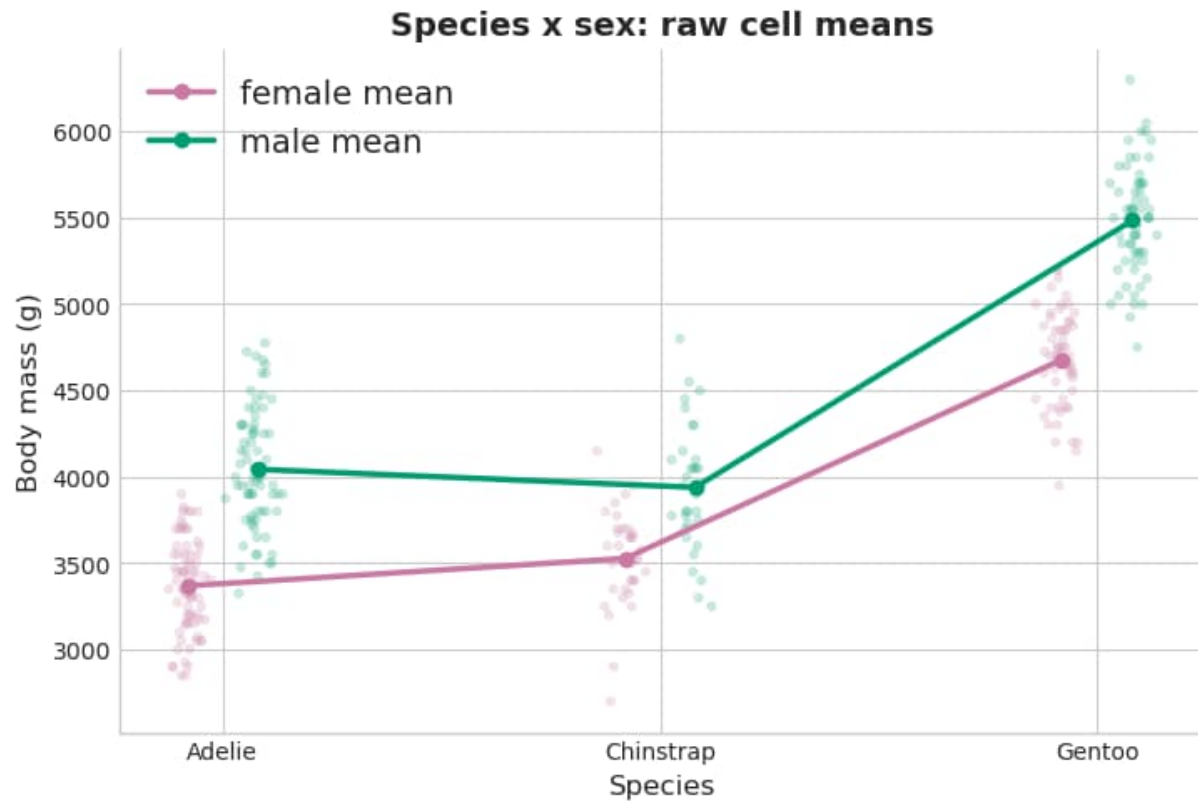
▼ Data variables:

sigma	(chain, draw)
species:sex	(chain, draw, species:sex_dim)
mu	(chain, draw, __obs__)

	mean	sd	hdi_3%	hdi_97%	ess_bulk	ess_tail	r_hat
species:sex[Adelie, female]	3368.289	36.171	3299.038	3436.406	6328.0	3365.0	1.0
species:sex[Adelie, male]	4043.687	36.972	3972.448	4111.917	6481.0	3086.0	1.0
species:sex[Chinstrap, female]	3526.981	54.733	3427.233	3630.533	5043.0	3009.0	1.0
species:sex[Chinstrap, male]	3938.505	53.760	3835.438	4040.343	4726.0	3107.0	1.0
species:sex[Gentoo, female]	4680.625	41.294	4605.525	4759.418	6112.0	3245.0	1.0
species:sex[Gentoo, male]	5485.206	40.831	5409.889	5561.664	5077.0	3319.0	1.0
sigma	310.445	12.107	287.822	333.716	4920.0	3082.0	1.0

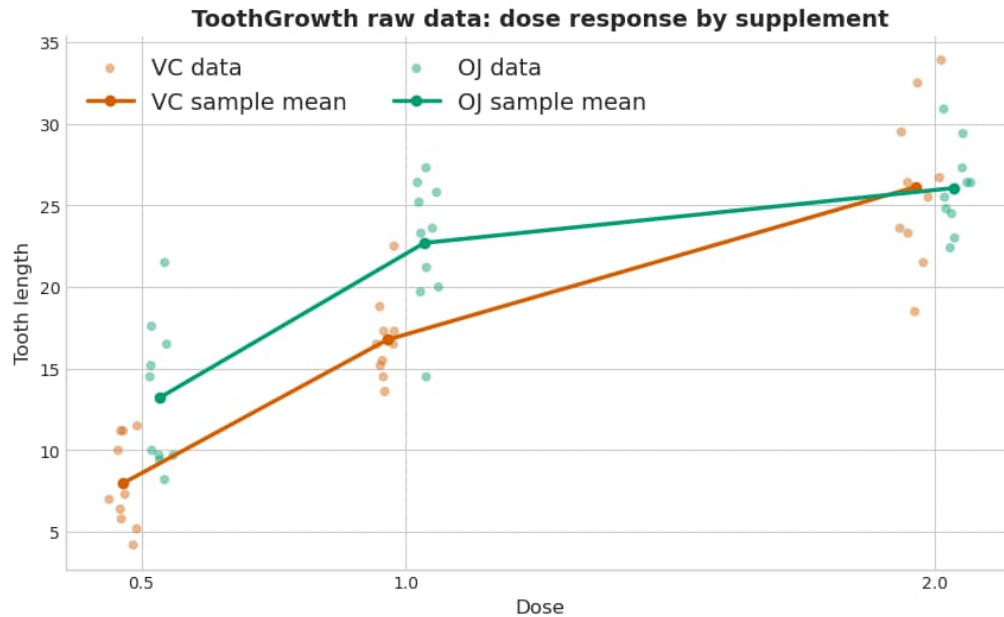
The interaction is visible by eye

- Sex difference
 - smaller in Chinstrap
 - And larger in Gentoo



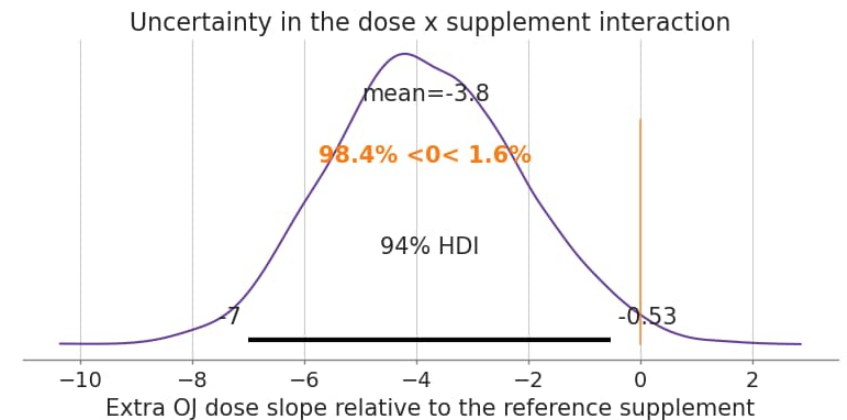
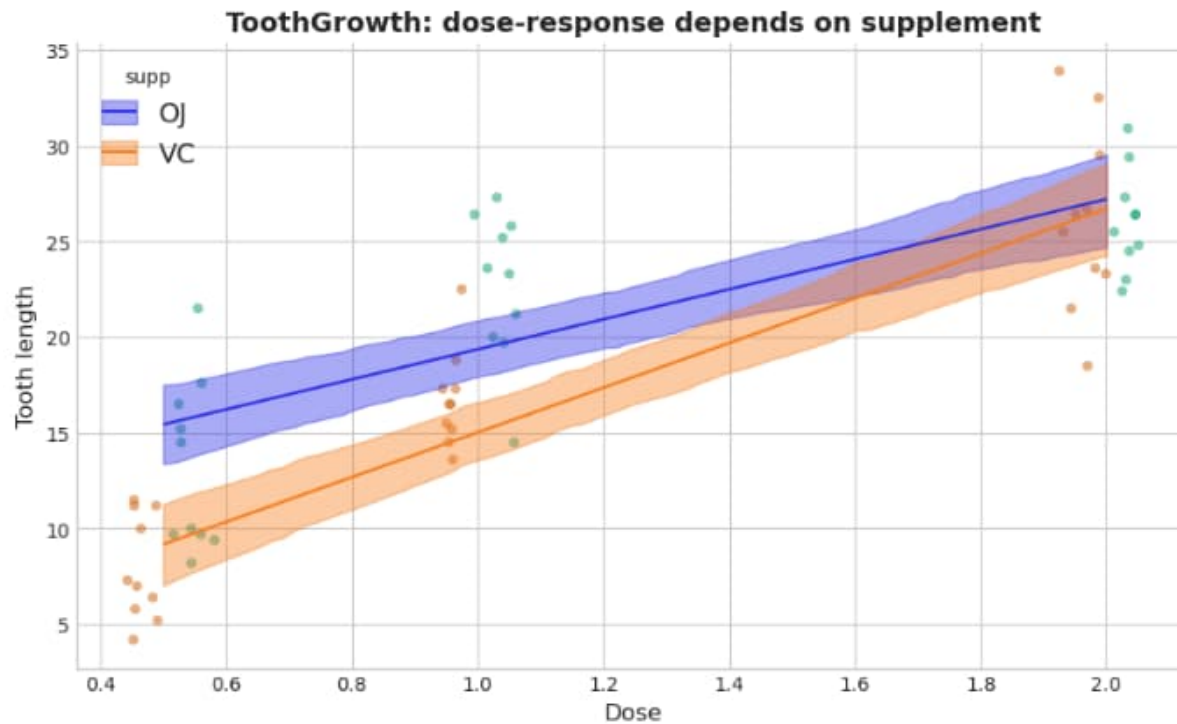
Tooth growth data shows interaction of continuous variables

- The model mixes categorical and continuous variables



```
bmb.Model(  
  "len ~ dose * supp",  
  data=tooth,  
)
```

Although posterior plot suggests categorical might work better

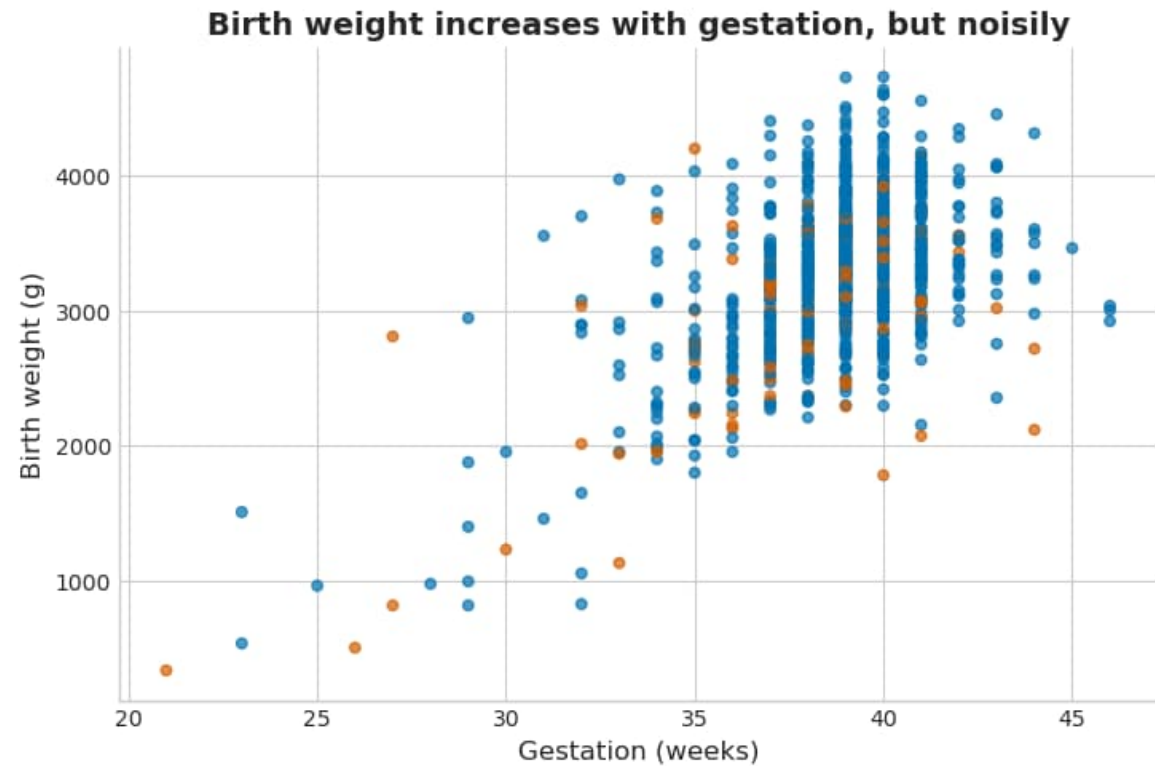


9.6 Categorical vs continuous predictors

What is the difference between using a categorical and a continuous predictor

An important biomedical question

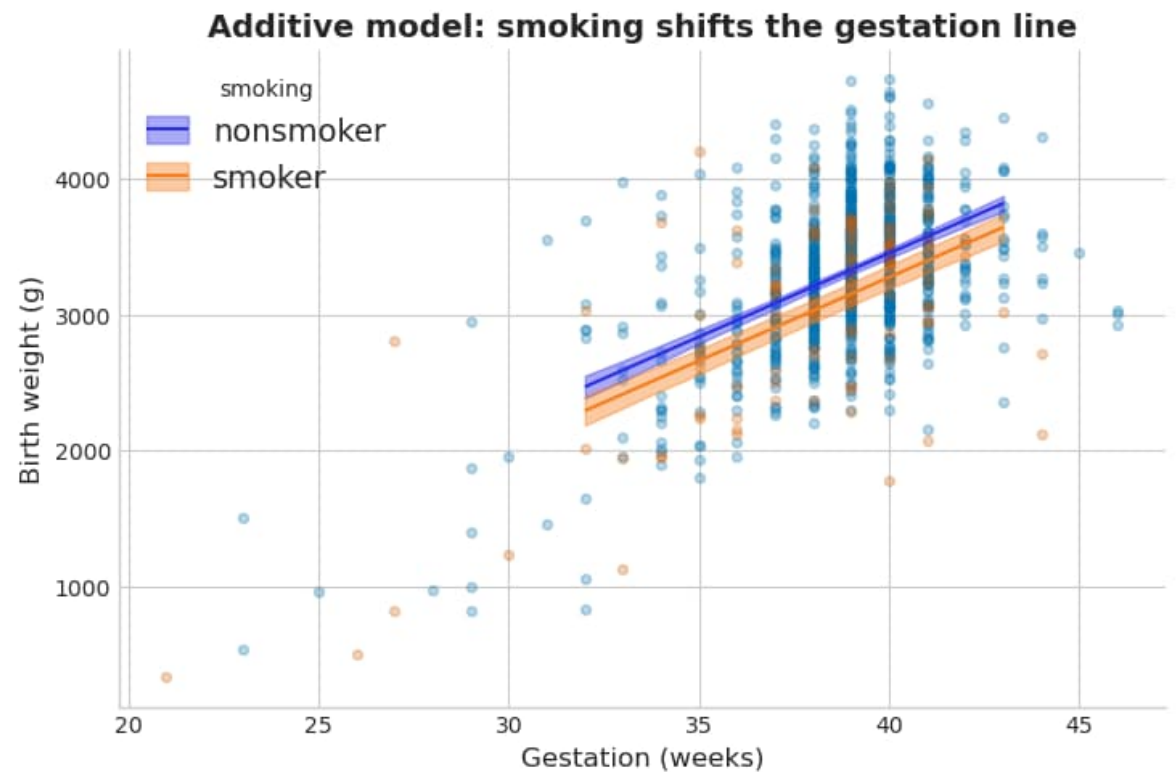
- Does smoking affect birth weight?



Fit an additive model and see a possible difference

- Smoking introduces a decrease of 175 g (HDI: -260 to 50)

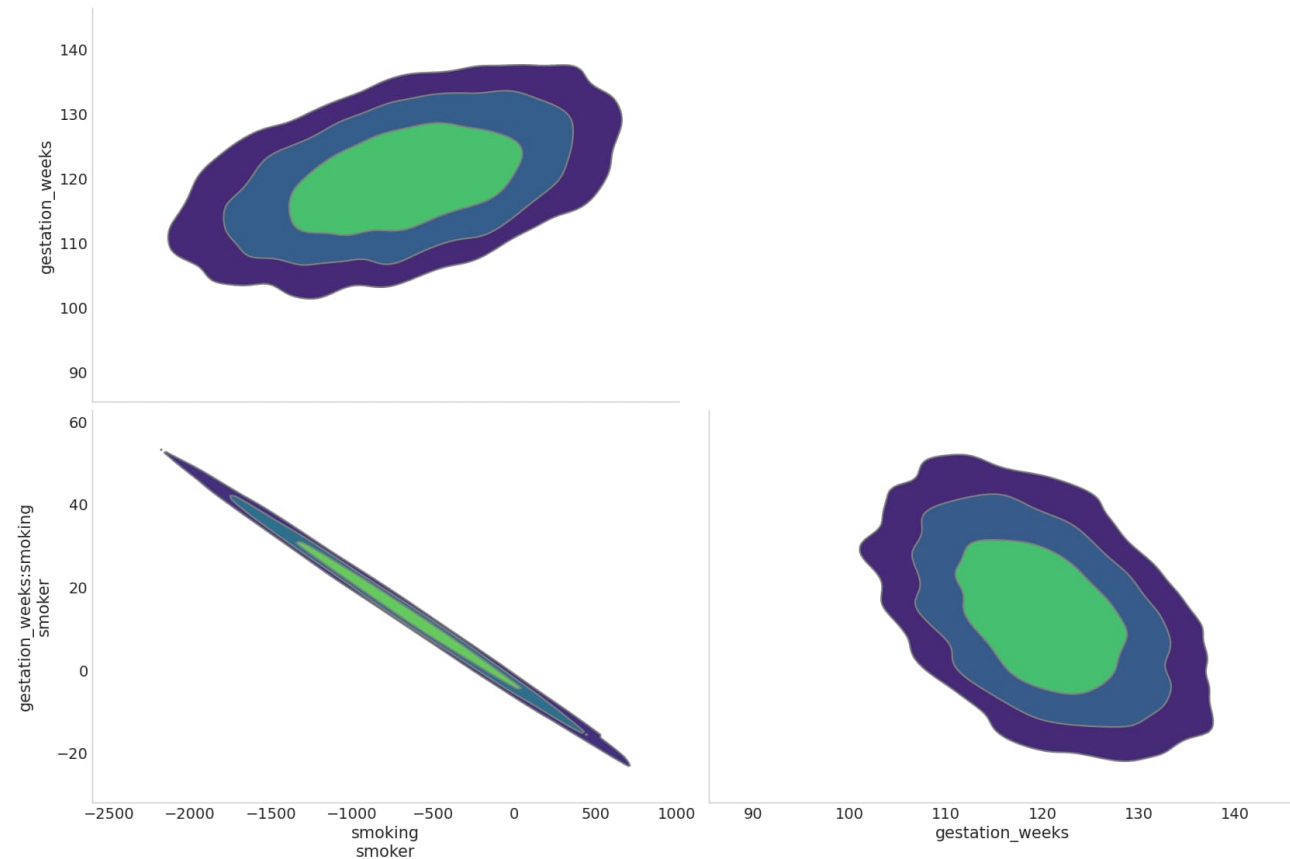
	mean	sd	hdi_3%	hdi_97%
Intercept	-1452.914	237.598	-1891.673	-1013.963
gestation_weeks	122.558	6.107	111.409	134.027
smoking[smoker]	-175.865	47.776	-261.325	-83.772
sigma	490.352	10.710	470.265	510.394



An interaction model is also problematic

- We can't tell if there is a small shift or an interaction

	mean	sd	hdi_3%	hdi_97%
Intercept	-1338.324	272.939	-1853.683	-820.826
gestation_weeks	119.598	7.027	106.297	132.862
smoking[smoker]	-690.403	585.954	-1760.188	441.184
gestation_weeks:smoking[smoker]	13.464	15.306	-16.700	41.071
sigma	490.688	11.322	469.995	512.822

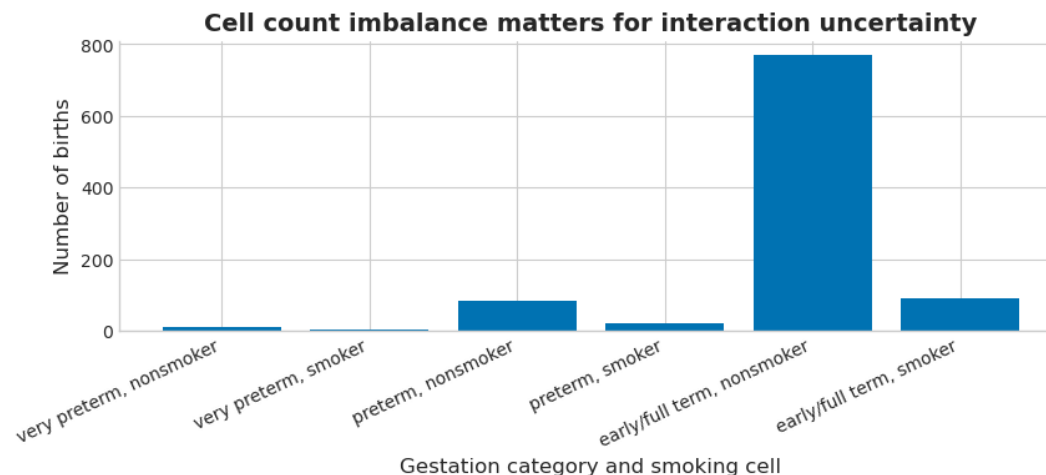


Maybe gestation should not be a continuous variable

- Is a difference of a week the same everywhere?
 - A linear model assumes that:
 - One week at term has the same effect as one week in month 7
- We can create categories
- Data will be unbalanced

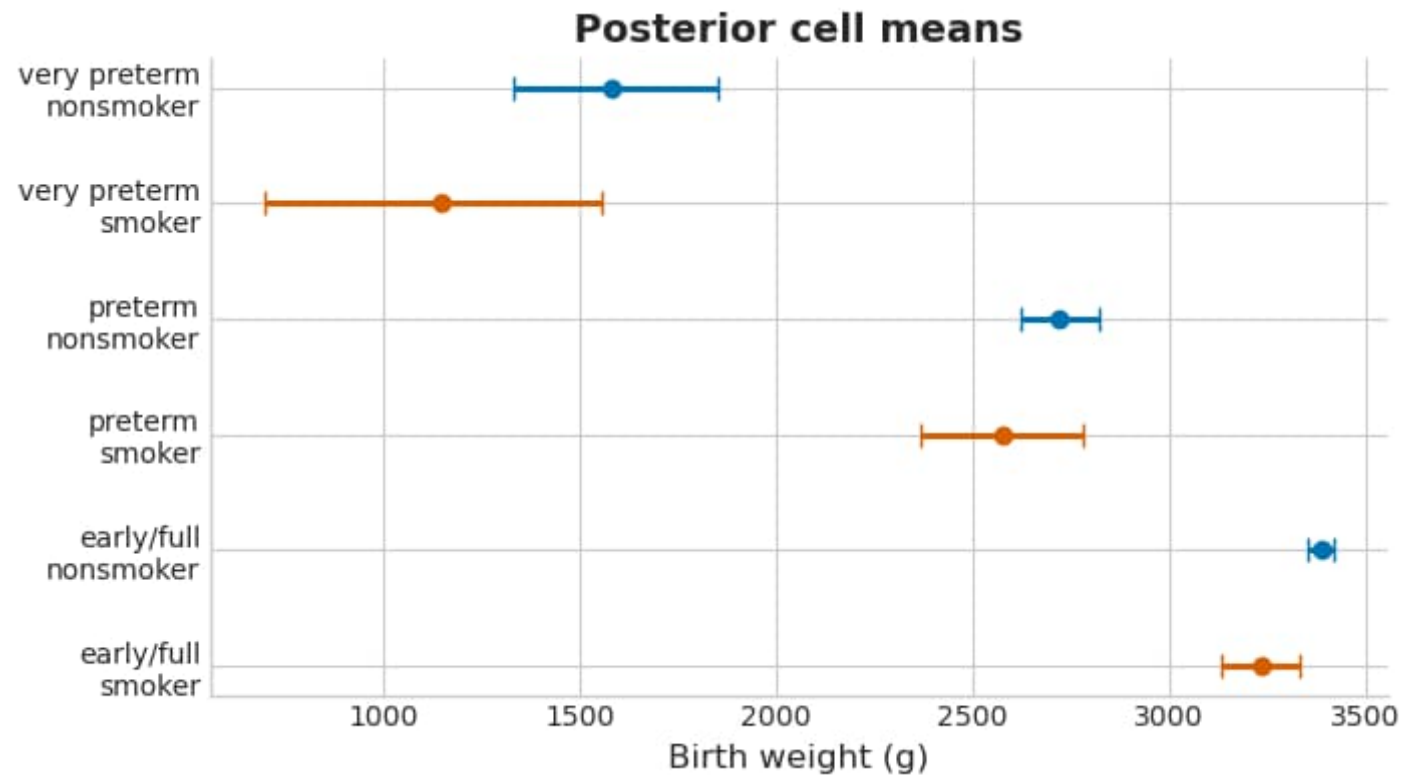
```
bmb.Model(  
  "birth_weight ~ 0 + gestation_category:smoking",  
  data=births_model_df,  
)
```

Category	Weeks
Very preterm	<32
Preterm	32 to <37
Early/full term	>=37



Effects may vary with gestational term

- HDIs suggest we don't have enough data
 - Should we repeat the study in a country with more smoking?
- Categorical coding clarifies the picture
 - In this case



9.6 Model effect sizes

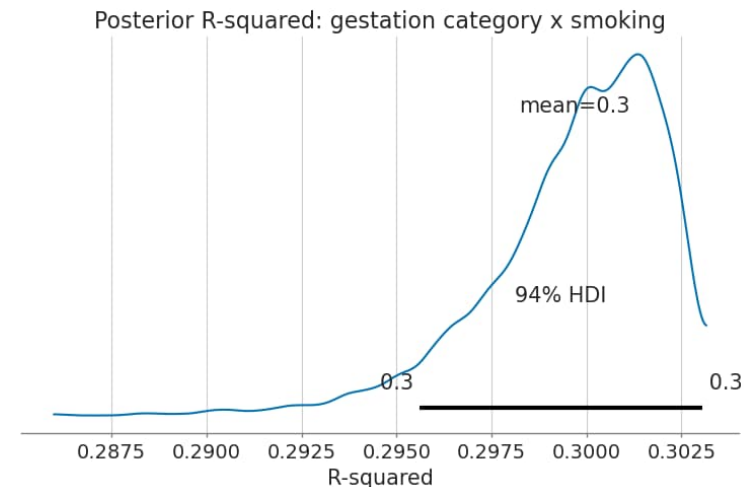
How much explanatory power does the model have?

R^2 is the variance explained by the regression

- The total variance is $\text{var}(y)$
 - Decomposed into:
 - The differences between the μ_i
 - What we know about y_i from knowing $x_{k,i}$
 - The data variability σ
 - These add because
 - Differences in μ_i are independent of likelihood noise

$$\text{var}(y) = \text{var}(\mu_i) + \sigma^2$$

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$



Frequentist R^2 is not an estimate of Bayesian R^2

- MS is the frequentist estimate of variance
 - SS is not

I said this last lecture and it is not right!!

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} \neq \frac{MS_{\text{reg}}}{MS_{\text{tot}}}$$

~~$$R^2 = \frac{\text{var}(\hat{y}_i)}{\text{var}(y_i)}$$~~

$$SS_{\text{reg}} = \sum_{i=1}^N (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{tot}} = \sum_{i=1}^N (y_i - \bar{y}_i)^2$$

$$df_{\text{reg}} = k$$

$$df_{\text{tot}} = N - 1$$

$$MS_{\text{reg}} = \frac{SS_{\text{reg}}}{df_{\text{reg}}}$$

$$MS_{\text{tot}} = \frac{SS_{\text{tot}}}{df_{\text{tot}}}$$

Partial R^2 is the variance explained by one regression variable

- What would μ_i be if we left out one term?
- How much variance is explained by including that term

$$\mu_i^{x_1} = \beta_0 + \beta_2 x_{2,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$R_{x_1}^2 = \frac{\text{var}(\mu_i^{x_1} - \mu_i)}{\text{var}(\mu_i^{x_1} - \mu_i) + \sigma^2}$$

$$\mu_i^{x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$R_{x_2}^2 = \frac{\text{var}(\mu_i^{x_2} - \mu_i)}{\text{var}(\mu_i^{x_2} - \mu_i) + \sigma^2}$$

$$\mu_i^{x_1 x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$$

$$R_{x_1 x_2}^2 = \frac{\text{var}(\mu_i^{x_1 x_2} - \mu_i)}{\text{var}(\mu_i^{x_1 x_2} - \mu_i) + \sigma^2}$$